

Problem: Given that $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$.

prove that (i) $\lim_{x \rightarrow c} [f(x) + g(x)] = L + M$.

(ii) $\lim_{x \rightarrow c} (f \cdot g)(x) = L \cdot M$

(iii) $\lim_{x \rightarrow c} \left(\frac{1}{g}\right)(x) = \frac{1}{M}$ if $M \neq 0$.

Solⁿ (i)

let $\epsilon > 0$, we want to find a positive number δ such that

$$|f(x) + g(x) - (L + M)| < \epsilon \text{ whenever } 0 < |x - c| < \delta.$$

$$\text{Now } |f(x) + g(x) - L - M|$$

$$= |f(x) - L + (g(x) - M)|$$

$$\leq |f(x) - L| + |g(x) - M|$$

Since $\lim_{x \rightarrow c} f(x) = L$, $\exists \delta_1 > 0$ such that

$$|f(x) - L| < \frac{\epsilon}{2} \text{ whenever } 0 < |x - c| < \delta_1.$$

again, $\lim_{x \rightarrow c} g(x) = M$, $\exists \delta_2 > 0$ s.t.

$$|g(x) - M| < \frac{\epsilon}{2} \text{ whenever } 0 < |x - c| < \delta_2.$$

$$\text{Let } \delta = \min\{\delta_1, \delta_2\}$$

If $0 < |x - c| < \delta$ then $|x - c| < \delta_1$, so $|f(x) - L| < \frac{\epsilon}{2}$ and

so $|f(x) - L| < \frac{\epsilon}{2}$ and

$$|x - c| < \delta_2, \text{ so } |g(x) - M| < \frac{\epsilon}{2}.$$

$$\text{Therefore, } |f(x) + g(x) - (L + M)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

$$\text{i.e. } \lim_{x \rightarrow c} (f(x) + g(x)) = L + M.$$

How to prove that $\lim_{x \rightarrow c} f(x) \neq L$. [Showing L is not a limit]

Sol. We want to prove that $\lim_{x \rightarrow c} f(x) \neq L$ by providing an $\epsilon > 0$ such that no possible $\delta > 0$ satisfies the condition

$$|f(x) - L| < \epsilon \text{ whenever } 0 < |x - c| < \delta.$$

For each $\delta > 0$ $\exists x$ such that

$$0 < |x - c| < \delta \text{ and } |f(x) - L| \geq \epsilon. \quad [\text{using negation}]$$

Problem: Prove that $\lim_{x \rightarrow 3} (x^2 + 4) \neq 5$.

Sol. Using negation:

$\exists \epsilon > 0$ such that $\forall \delta > 0$, there is an x with $0 < |x - 3| < \delta$ but $|(x^2 + 4) - 5| \geq \epsilon$.

Choose $\epsilon = 1$

For any $\delta > 0$ select $x = 3 + \frac{\delta}{2}$

$$\text{so, } 0 < |x - 3| = \left| 3 + \frac{\delta}{2} - 3 \right| = \frac{\delta}{2} < \delta.$$

$$\begin{aligned} \text{Now, } |(x^2 + 4) - 5| &= \left| \left(3 + \frac{\delta}{2} \right)^2 + 4 - 5 \right| = \left| 9 + 3\delta + \frac{\delta^2}{4} - 1 \right| \\ &= \left| 8 + 3\delta + \frac{\delta^2}{4} \right| \end{aligned}$$

$$\text{since } \left| 8 + 3\delta + \frac{\delta^2}{4} \right| > 8 > 1 = \epsilon.$$

$$\text{Thus, } |(x^2 + 4) - 5| \geq \epsilon.$$

□

Some Extra problems:

Using ϵ - δ definition:

1. $\lim_{x \rightarrow 4} (9-x) = 5$

2. $\lim_{x \rightarrow 1} f(x) = 1$ if $f(x) = \begin{cases} x^2, & x \neq 1 \\ 2, & x = 1 \end{cases}$

3. $\lim_{x \rightarrow 0} \sqrt{4-x} = 2$

4. $f(x) = 3-2x$, $c = 3$, $\epsilon = 0.02$

5. ~~$\lim_{x \rightarrow 1} f(x) = 2$ if $f(x) = \begin{cases} 4-2x, & x < 1 \\ 6x-4, & x > 1 \end{cases}$~~

5. $\lim_{x \rightarrow -2} f(x) = 4$ if $f(x) = \begin{cases} x^2, & x \neq -2 \\ 1, & x = -2 \end{cases}$

6. let $h(x) = \begin{cases} (x^2+1)^2, & x \neq 2 \\ 2, & x = 2 \end{cases}$

Show that $\lim_{x \rightarrow 2} h(x) \neq 4$

Continuity

Now we defined a function

$$f(x) = \begin{cases} x^2, & x < 0 \\ x, & x \geq 0 \end{cases}$$

Since when $x=0$, $f(0)=0$

$$\text{Now L.H.L.} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x^2 = 0$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0$$

So, L.H.L. = R.H.L. which implies limit exists and we have a limiting value at $x=0$.

Here we also define $f(0)=0$.

We use here extra information which is ^{function is} ~~is~~ defined at $x=0$ when extra condition also satisfy then we say the function is continuous at that point. Now we write the definition.

Definition of Continuity:

A function f is said to be continuous at $x=a$ provided the following conditions are satisfied:

① $f(a)$ is defined

② $\lim_{x \rightarrow a} f(x)$ exists

and ③ $\lim_{x \rightarrow a} f(x) = f(a)$.



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Example: Given a function,

$$f(x) = \begin{cases} x^2 + 1, & x > 0 \\ 1, & x = 0 \\ x + 1, & x < 0 \end{cases}$$

Show that the function $f(x)$ is continuous at $x = 0$.

Sol. at $x = 0$:

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x + 1) = 1$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x^2 + 1 = 1$$

L.H.L. = R.H.L. i.e. limit exist.

Since $f(0) = 1$ i.e. $f(0)$ is defined.

$$\text{Thus, } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = 1 = f(0).$$

Therefore, $f(x)$ is continuous at $x = 0$.

✕

Problem: Given that $f(x) = \begin{cases} e^{-|x|/2} & , -1 < x \leq 0 \\ \tilde{x} & , 0 < x < 2 \end{cases}$

Does the above function continuous at $x=0$?

Solⁿ First we want to show that function is defined at $x=0$

$$f(0) = e^{-|0|/2} = e^0 = 1$$

Now for limit exists.

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} e^{-|x|/2} = \lim_{x \rightarrow 0^-} e^{-x/2} = 1$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \tilde{x} = 0$$

So, L.H.L. $\neq$ R.H.L. i.e. limit does not exist.

Therefore, the above function is not continuous at $x=0$.

Properties of Continuity

If the functions f and g are continuous at a , then

1. $f \pm g$ is continuous at a .

2. $f \cdot g$ is continuous at a .

3. f/g is continuous at a ~~if~~ ^{and} $g(a) \neq 0$.

4. If $\lim_{x \rightarrow a} g(x) = L$, and if f is continuous then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

* Consider $g(x) = x+1$, and $f(x) = \sqrt{x}$ is

continuous for $x \geq 0$. If $\lim_{x \rightarrow 2} g(x) = 5$ then
we find the limit

$$\lim_{x \rightarrow 2} f(g(x)) = ?$$

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Some Extra Problems:

1. Given that $f(x) = \begin{cases} x^2 - 6x + 10, & x \geq 2 \\ 2 - x, & x < 2 \end{cases}$,

Test the continuity at $x = 2$.

2. Given that $g(x) = \begin{cases} 2x + 3, & x \leq 4 \\ 7 + \frac{16}{x}, & x > 4 \end{cases}$.

Does the function, $g(x)$, continuous at $x = 4$?

3. ~~Given~~ Test the continuity of the following function at $x = 0$:

$$f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} - 1} & ; \text{ if } x \neq 0 \\ 1 & , \text{ if } x = 0 \end{cases}$$

4. Does the following function continuous at $x = 0$?

$$f(x) = \begin{cases} x \cos(1/x) & , \text{ if } x \neq 0 \\ 0 & , \text{ if } x = 0 \end{cases}$$